

Lab Module Rulebook for Inpatient Medical Coding

Pipeline:

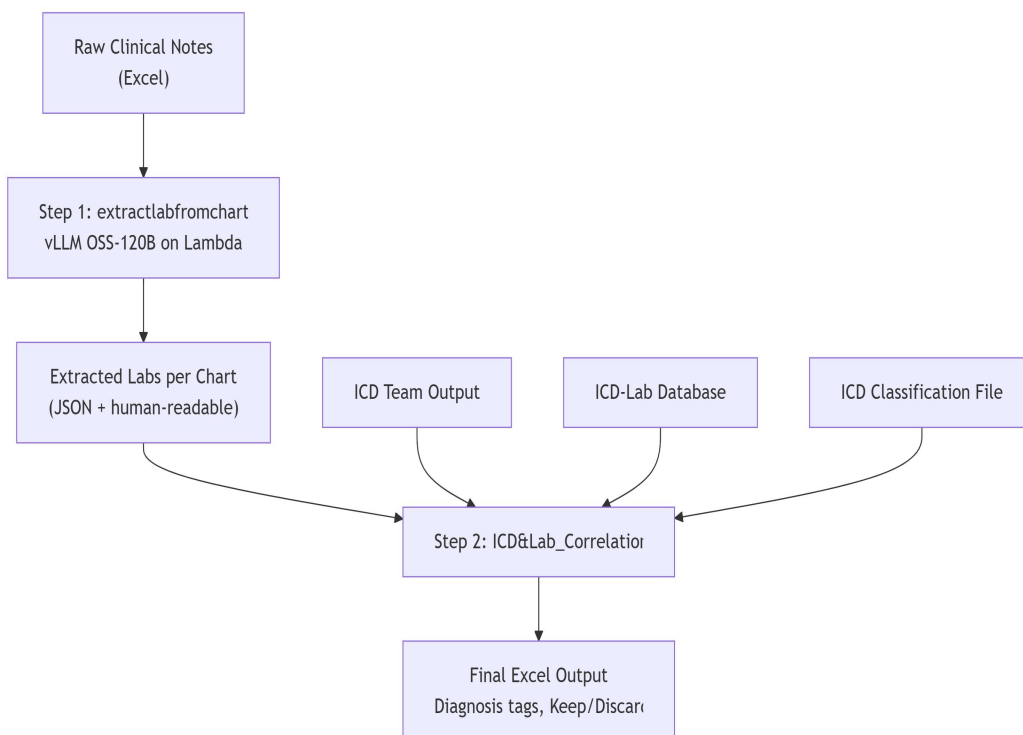
`extractlabfromchart.ipynb` – Lab/diagnostic extraction using vLLM

`ICD&Lab_Correlation_Code.ipynb` – ICD-lab correlation and coding support

1. Overview & Architecture

The pipeline processes inpatient clinical notes to extract laboratory results, vital signs, imaging, microbiology, and other diagnostic parameters. It then correlates these findings with ICD codes to support medical coding decisions.

System diagram:



2. Step 1 – Lab/Diagnostic Extraction

2.1 Input Specifications

Column name	Type	Description	Example
Chart_No	string	Unique patient identifier	200128981
Visit	string	Visit identifier	2024-01-15-001
DOS	date	Date of service	01/15/2024
Visit_Time	string	Time of visit	14:30
LAB RESULTS	text	Lab results section	Free text
VITAL SIGNS	text	Vital signs section	Free text
DIAGNOSTIC RESULTS	text	Diagnostic tests	Free text
IMPRESSION	text	Clinical impression	Free text
ASSESSMENT AND PLAN	text	A&P section	Free text
OBJECTIVE	text	Objective findings	Free text
ORDERS	text	Ordered tests	Free text
PHYSICAL EXAM	text	Physical exam	Free text

2.2 Extraction Rules

2.2.1 Text Merging and Row Filtering

- **Merge rule:** Combine all text columns with section headers, e.g.
--- LAB RESULTS ---\nWBC 12.5 ...
- **Skip empty rows:** All text columns empty → skip.
- **Skip orders-only rows:**
 - Contains Order Date, Ordered, InProcess, Scheduled **AND**
 - No numeric result patterns (e.g., \d+\.\d+, \d+\s*(L\)) → skip.

2.2.2 Five Parallel Extractors

Each row spawns 5 concurrent LLM extractors:

1. **Lab results** (lab name, value, flag, datetime)
2. **Vital signs** (5 parameters: O2 Sat, RR, BP systolic/diastolic, HR)
3. **Imaging** (X-ray, CT, MRI, Echo, ECG, etc.)
4. **Other** (oxygen flow, ejection fraction, GCS, NIHSS, PEG/chest tube)
5. **Microbiology** (cultures, PCR, Gram stain)

Configuration

- vLLM endpoint: http://localhost:8000/v1/chat/completions
- Model: /lambda/nfs/NDS1/03_MedicalCoding/OSS_120B
- Temperature: 0.0 (deterministic)
- Max tokens: 4096
- Timeout: 300 seconds
- Row workers: 6 → up to 30 concurrent LLM calls.

2.2.3 Lab Extraction Details

Flag mapping:

Input	Output label
(L), (Low), Low	L
(H), (High), High	H
(HH), Critical High	HH
(LL), Critical Low	LL
No flag	NORMAL

Standardised lab names:

Raw input	Standardised name
Hgb, Hemoglobin	Hemoglobin
WBC, White Blood Cells	WBC
BNP, Brain Natriuretic Peptide	BNP
Troponin I HS	Troponin I
HbA1c, Hemoglobin A1c	HbA1c

Output JSON schema:

```
{  "labs": [{
    "lab_name": "Hemoglobin",
    "lab_value": 14.2,
    "lab_label": "NORMAL",
    "datetime": "2024-01-15 08:30"
  } ]}
```

2.2.4 Vital Signs Rules

Only 5 parameters are extracted, **numeric values only** (no units/flags).
Excluded: Temperature, BMI, MAP, pain scores.

Datetime resolution:

- MM/DD HH:MM + DOS year → YYYY-MM-DD HH:MM
- HH:MM only + DOS date → YYYY-MM-DD HH:MM

2.2.5 Imaging & Categorical Results

Modality	Standardised name	Value interpretation
CXR, Chest X-ray	X-ray	Positive (abnormal), Negative (normal), Ordered
CT, CTA	CT	
MRI, MRA	MRI	
Echo, TTE, TEE	Echocardiogram	
EKG, ECG	ECG	

2.2.6 Microbiology Rules

Culture type mapping

- blood → Blood Culture
- urine → Urine Culture
- sputum → Sputum Culture

- pcr → PCR
- gram → Gram Stain

Skip if result contains InProcess, Preliminary, Ordered (no final result).

2.2.7 Special Items

Item	lab_name	lab_value
Oxygen via nasal cannula	Oxygen via nasal cannula	flow rate (e.g., 8 L/min)
Chest tube	Chest Tube	Present / Ordered

2.2.8 Post-processing (Deduplication & Grouping)

- **Deduplication key:** (lab_name, datetime)
Duplicate values at same timestamp are merged; visit list is updated.
- **Sorting:**
 - Global sort by datetime ascending.
 - Group by lab_name.
 - Within group, sort by datetime.
 - Groups ordered by minimum datetime of that lab.
- **Output:**
 - Extracted_Labs (JSON) – complete structured data.
 - Extracted_Labs_Sorted (string) – human-readable grouped format.

3. Step 2 – ICD-Lab Correlation

3.1 Input Specifications

File	Source	Description
ICD_Team_Output.xlsx	ICD team	ICD codes per chart
IP_data_extraction_final_01June2026.xlsx	Raw data	Original clinical notes (for ambiguity detection)
Lab_extracted_per_chart_complete_oss.xlsx	Step 1 output	Extracted labs per chart
Acute_or_chronic_database.xlsx	Classification file	ICD-10 category mapping (acute/chronic)
Final_Lab_database.csv	Lab database	ICD-lab associations with normal ranges

3.2 Lab Name Normalisation

Steps:

1. Lowercase
2. Remove special characters (keep alphanumeric, spaces, /, ., +, -)
3. Remove suffixes: lvl, level, value, result, relative, absolute
4. Apply synonym dictionary (excerpt shown below)

Raw	Normalised
wbc, white blood cells	white blood cells
hgb, hb, hemoglobin	hemoglobin
plt, platelets	platelets
na, sodium lvl	sodium
bnp, brain natriuretic peptide	bnp
troponin i hs	troponin

3.3 Normal Range Parsing

Patterns:

Pattern	Example	Output (low, high)
low-high	4.5-11.0	(4.5, 11.0)
< value	<10	(None, 10)
> value	>100	(100, None)
≤ value	≤0.5	(None, 0.5)

Sub-range extraction

WBC: 4.5-11.0, Hgb: 13.5-17.5 →
 { "white blood cells": (4.5,11.0), "hemoglobin": (13.5,17.5) }

3.4 Panel Expansion

Panel name	Expanded components
complete blood count	WBC, RBC, Hemoglobin, Hematocrit, Platelets, RDW, MCV, MCH, MCHC, MPV
basic metabolic panel	Sodium, Potassium, Chloride, CO ₂ , Glucose, BUN, Creatinine, Calcium
comprehensive metabolic panel	BMP + Albumin, Total Protein, AST, ALT, Alkaline Phosphatase, Total Bilirubin
liver function tests	ALT, AST, Alkaline Phosphatase, Total Bilirubin, Albumin, Total Protein
lipid panel	Total Cholesterol, LDL, HDL, Triglycerides
coagulation panel	PT, INR, PTT

3.5 Special ICD Handling (R, V, W, X, Y, Z)

If the ICD code starts with R, V, W, X, Y, or Z:

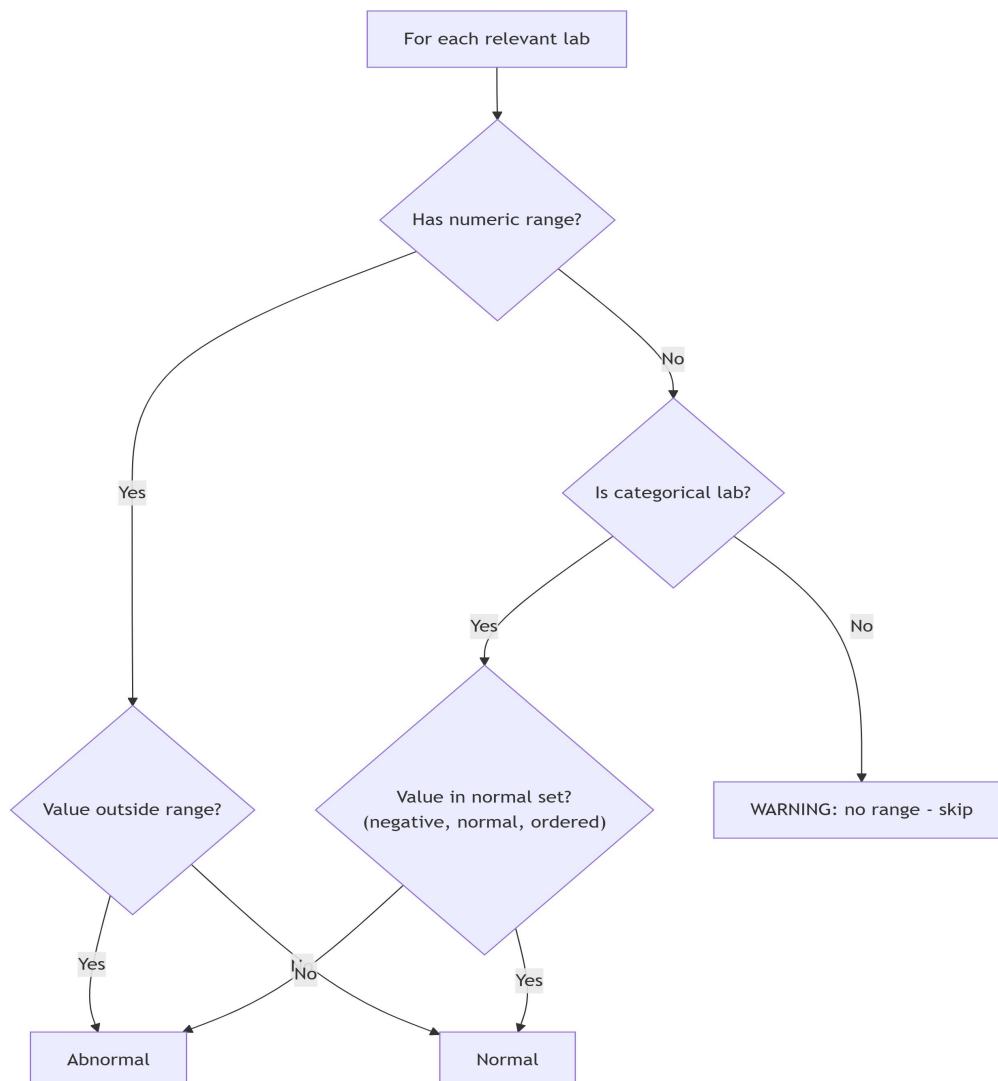
- Add to **Keep** list automatically.
- Skip lab processing.
- Do not generate a diagnosis tag or abnormal summary.
- Still include in `ICD_team_codes` column.

3.6 Ambiguous Wording Detection

Keywords: previous condition, previously treated, treated, resolved.

- If any keyword appears in the chart text → the whole chart is flagged as ambiguous.
- For a condition with **abnormal labs**, if the condition name (or ICD) appears within **200 characters** of an ambiguous keyword → the case is added to the **Query** output, requiring manual review.

3.7 Abnormal / Normal Classification



Categorical lab set (partial):

x-ray, ct, mri, ultrasound, echocardiogram, ecg, blood culture, urine culture, pcr, cardiac cath, carotid doppler.

3.8 ICD Category Classification (Acute/Chronic)

- Remove decimal point from ICD code, take first **3 characters** as prefix.

- Match prefix against ranges in `Acute_or_chronic_database.xlsx` (Sheet1 first, then Sheet2).
- Default: `Unknown`.

4. Output Specifications

4.1 Step 1 Output – `extracted_per_chart_final.xlsx`

Column	Type	Description
<code>Chart_No</code>	string	Chart identifier
<code>Extracted_Labs</code>	JSON	Full list of lab entries (structured)
<code>Extracted_Labs_Sorted</code>	string	Human-readable, deduplicated, grouped by lab name

JSON structure example:

```
{ "labs": [ { "lab_name": "Hemoglobin", "lab_value": 14.2, "lab_label": "NORMAL", "datetime": "2024-01-15 08:30", "visit": "2024-01-15-001", "visit_dos": "2024-01-15" } ] }
```

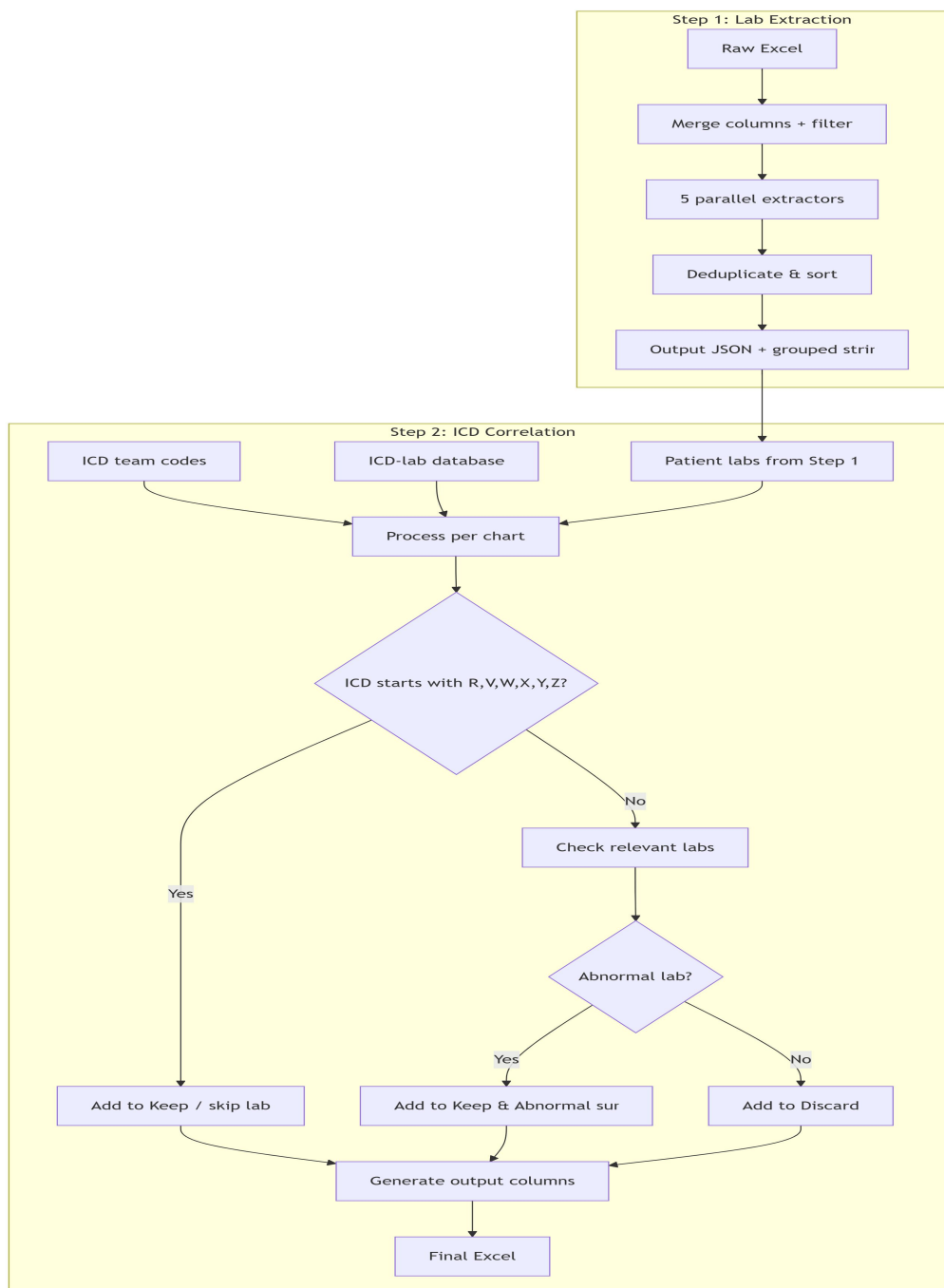
4.2 Step 2 Output – `ICD_Lab_Correlation_&_Query_Output_[date].xlsx`

Column	Description
<code>Chart_number</code>	Chart identifier
<code>ICD_team_codes</code>	All ICD codes from ICD team (one per line)
<code>Diagnosis_tag</code>	Full diagnostic tag: ICD:Diagnosis { Normal(...) Abnormal(...) Category }
<code>Abnormal_Labs_Summary</code>	Only ICDs with at least one abnormal lab
<code>Keep</code>	ICDs with abnormal labs or R/V/W/X/Y/Z codes
<code>Discard</code>	ICDs with no abnormal labs (normal labs only)
<code>Query</code>	Cases with ambiguous wording + abnormal labs → manual review needed

Example `Diagnosis_tag`:

```
E11.42:Type 2 diabetes mellitus with diabetic polyneuropathy { Normal(Glucose:110, HbA1c:6.5) | Abnormal() | Chronic }I10:Essential hypertension { Normal() | Abnormal(BP Systolic:165, BP Diastolic:95) | Chronic }
```

5. Complete Workflow Chart



6. Interpretation Guide

6.1 Understanding `Extracted_Labs_Sorted`

Each line represents one deduplicated lab measurement.
Labs are grouped by name, and groups are sorted by the earliest datetime.

6.2 Understanding `Diagnosis_tag`

- `Normal(...)` – all measured labs for that ICD are within range.

- `Abnormal(...)` – at least one lab is outside range (flag H/HH/L/LL).
- `Category` – `Acute` Or `Chronic` (from classification file).

6.3 Using Keep vs Discard

- **Keep** = likely active, relevant for coding.
- **Discard** = probably historical or resolved; may be omitted unless required by payer rules.

6.4 Interpreting Query

Query appears when:

1. The chart contains ambiguous wording (`previous`, `treated`, `resolved`).
2. A condition has abnormal labs.
3. The condition name (or its ICD) appears near the ambiguous phrase.

Action: Manually review the chart to decide if the condition is still active or truly resolved.

7. Error Handling & Troubleshooting

Warning / Error	Probable cause	Solution
WARNING: No numeric range for ICD X, lab Y	Lab not in database for this ICD	Add range to <code>Final_Lab_database.csv</code> or verify ICD-lab association.
Warning: Unknown categorical value '...'	Unexpected value (e.g., <code>Pending</code>)	Update <code>CATEGORICAL_LAB_NAMES</code> or treat as abnormal.
Warning: Could not parse labs for chart X	Malformed JSON in Step 1 output	Re-run Step 1 for that chart.
[LLM ERROR]	vLLM service not reachable	Check Lambda instance, endpoint, model path.
ValueError: Cannot find ICD column	Column name mismatch in ICD team file	Use <code>unique_icd</code> or <code>ICD_Codes</code> accordingly.

8. Appendices

8.1 Configuration Parameters

Step 1 (in `extractlabfromchart.ipynb`)

```
VLLM_URL = "http://localhost:8000/v1/chat/completions"
MODEL_NAME = "/lambda/nfs/NDS1/03_MedicalCoding/OSS_120B"
```

```
TEMPERATURE = 0.0
```

```
row_workers = 6
```

Step 2 (in `ICD&Lab_Correlation_Code.ipynb`)

```

AMBIGUOUS_KEYWORDS = ['previous condition', 'previously treated', 'treated', 'resolved']
CATEGORICAL_LAB_NAMES = {"x-ray", "ct", "mri", "ultrasound", "echocardiogram", "ecg", "blood culture", "urine culture", "pcr"}

```

8.2 File Paths (customise for your environment)

Step 1

```

input_file = "/lambda/nfs/NDS1/New_LabIP_OSS/IP_data_extraction_final_01June2026.xlsx"
output_file = "/lambda/nfs/NDS1/New_LabIP_OSS/Output/extracted_per_chart_final.xlsx"

```

Step 2

```

EXTRACTION_FILE = 'ICD_Team_Output.xlsx'
CHART_FILE = 'IP_data_extraction_final_01June2026.xlsx'
LABS_FILE = 'Lab_extracted_per_chart_complete_oss.xlsx'
ICD_CLASS_FILE = 'Acute_or_chronic_database.xlsx'
LAB_DATABASE_FILE = 'Final_Lab_database.csv'

```

8.3 Synonym Dictionary Sample (for normalisation)

A full dictionary is available in the code; key entries include:

```

"wbc" → "white blood cells", "hgb" → "hemoglobin", "plt" → "platelets"
"na" → "sodium", "k" → "potassium", "bnp" → "bnp", "troponin i hs" → "troponin"

```

9. Example Walkthrough

Input (excerpt from clinical note)

LAB RESULTS:WBC 12.5 (H), Hgb 14.2, Sodium 135

IMPRESSION:Sepsis likely

Step 1 output (JSON)

```

{"labs":[
{"lab_name":"WBC","lab_value":12.5,"lab_label":"H","datetime":"2024-01-15 08:30"},
{"lab_name":"Hemoglobin","lab_value":14.2,"lab_label":"NORMAL","datetime":"2024-01-15 08:30"},
{"lab_name":"Sodium","lab_value":135,"lab_label":"NORMAL","datetime":"2024-01-15 08:30"}]}

```

Step 2 correlation

For ICD A41.9 (Sepsis), relevant labs: WBC, Hemoglobin, Sodium.

Normal ranges: WBC 4.5-11.0 → WBC 12.5 is abnormal; others normal.

Output:

```

A41.9:Sepsis, unspecified { Normal(Hemoglobin:14.2, Sodium:135) | Abnormal(WBC:12.5) | Acute }

```

→ Added to **Keep** and **Abnormal_Labs_Summary**.